

Strong Customer Authentication for Apple Pay on iPhone SE (3rd generation) with A15 Bionic run- ning iOS 18.4

Guidance

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Table of Contents

- 1. Introduction3**
- 2. Preparation Guidance4**
- 3. Identification4**
- 4. Operational Guidance4**
 - 4.1. Configure Passcode5**
 - 4.2. Check warranty status5**
 - 4.3. Configure Touch ID5**
 - 4.4. Update iOS5**
 - 4.5. Apple Pay5**
 - 4.6. Apple Cash6**
 - 4.7. Operational failures6**
 - 4.8. Security updates, announces and registering6**
- Annex A - Card issuer Security Objectives7**
- Annex B - Apple Server Security Objectives8**
- Annex C – User Security Objectives9**

1. Introduction

This document contains references to other documents providing guidance for security related topics specified in the Security Target.

Reference	Description
[AP]	Apple Pay Support https://support.apple.com/apple-pay
[APC]	Apple Cash Support https://support.apple.com/apple-cash
[APS]	Apple Platform Security, Dec 2024 https://support.apple.com/guide/security
[CHECK-SERIAL]	Check Your Service and Support Coverage (review your Apple warranty status) https://checkcoverage.apple.com
[DEVICE_ID]	PSD2 security certifications - Device Identity
[DISABLE]	If you forgot your iPhone passcode https://support.apple.com/118430
[INITCFG]	Set up your iPhone https://support.apple.com/105132
[IOSID]	Find the software version on your iPhone https://support.apple.com/109065
[IOSSLA]	A. Apple iOS Software License Agreement B. Apple Pay Supplemental Terms and Conditions https://www.apple.com/legal/sla/docs/iOS18_iPadOS18.pdf
[IOSUPDATE]	Update the iOS on your iPhone https://support.apple.com/118575
[IPHONEID]	Identify your iPhone model https://support.apple.com/108044
[PASSCODE]	Use a passcode with your iPhone https://support.apple.com/119586
[PERSONAL-SAFETY]	Personal Safety User Guide for Apple devices Set a unique passcode or password on devices https://support.apple.com/guide/personal-safety/welcome/1.0/web
[SEC-ANNOUNCE]	Registration form for Apple security-announce mailing list https://lists.apple.com/mailman/listinfo/security-announce/
[SEC-ISSUE]	Get help with security issues https://support.apple.com/111756
[SEC-REPORT]	Report a security or privacy vulnerability https://support.apple.com/102549
[SEC-UPDATE]	Apple security releases https://support.apple.com/100100
[SERIAL]	Find the serial number of your iPhone https://support.apple.com/108037
[TOUCHID_ABOUT]	About Touch ID advanced technology https://support.apple.com/105095
[TOUCHID_ISSUE]	If Touch ID isn't working on iPhone https://support.apple.com/101612
[TOUCHID]	Use Touch ID on iPhone https://support.apple.com/102528
[USER_GUIDE]	iPhone User Guide https://support.apple.com/guide/iphone/welcome/ios

2. Preparation Guidance

After unpacking and powering up the device for the first time or after a complete erase, the iOS device presents a set of questions to the user as outlined in [INITCFG]. In addition to the steps outlined in [INITCFG], the user is required to:

- Select the language to be used by the OS and applications, and the User's country or region
- Select the age range of the User of the device
- Agree to [IOSSLA], the iOS Software License Agreement (SLA)

As part of the initial configuration, the user is asked to configure a passcode and enroll into Touch ID biometric authentication.

After completion of the initial installation steps, the user shall¹ enroll into Apple Pay and can elect to enroll into Apple Cash (if available). The enrollment process is illustrated at [AP]. To enable Apple Cash, the guidance given at [APC] should be consulted.

3. Identification

Two guides [IPHONEID] and [IOSID] are provided for identifying the device model and the installed software:

The following identifiers correspond to the TOE:

- TOE: Strong Customer Authentication for Apple Pay, on iPhone SE (3rd generation) with A15 running iOS 18.4
- Device Model: iPhone SE (3rd generation)
- iOS version: iOS 18.4

The other components of the TOE are tied to the device and iOS versions listed above and are not configurable.

4. Operational Guidance

In addition to the initial configuration steps, various use cases and options are available for the security functions at runtime. All security related mechanisms are documented as follows.

In general, all security features of iOS devices including authentication, system updates, Apple Pay, and Apple Cash are documented in [APS]. In addition, specific user guidance is given in the documents referenced in subsequent sections of this document.

Apple provides a high-level document covering the iOS Software License and Agreement [IOSSLA], including supplemental terms and conditions for the use of Apple Pay services (Apple Pay and Apple Cash).

The only user role applicable to the TOE is the end user of the device, therefore, the functions and privileges described in this document apply only to the end user.

¹ Enrolling into Apple Pay is not a mandatory step of the device setup process; however, it is required to install the TOE in its evaluated configuration

There is only one operational mode in the certified configuration of the TOE.

4.1. Configure Passcode

Managing the passcode is provided with the configuration user interface specified in [PASSCODE]. The guidance provides details about adding, changing, and removing a passcode.

To prevent anyone except the user from using their devices and accessing their information, the user should set a unique passcode or password that only the user knows. The Personal Safety User Guide [PERSONAL-SAFETY] and the "Privacy and security" section of [USER_GUIDE] provide guidance on setting up a passcode or password on devices.

To discourage brute force passcode attacks, there are escalating time delays after the entry of an invalid passcode, as specified in the Security Target (FIA_AFL.1/Delay Authentication failure handling).

4.2. Check warranty status

The documents [SERIAL] and [CHECK-SERIAL] instruct the user how to check the warranty status of their Apple devices.

4.3. Configure Touch ID

iOS allows the configuration of Touch ID by allowing users to enroll one or more fingerprints and managing previously enrolled fingerprints, including their removal. All configuration steps pertaining to these actions are given in [TOUCHID] and the "Privacy and security" section of [USER_GUIDE].

[TOUCHID] and [TOUCHID_ABOUT] provide information about how Touch ID is used to unlock the device and during Apple Pay and Apple Cash transactions.

4.4. Update iOS

The iOS operating system can be updated following the steps provided in [IOSUPDATE].

iOS updates include all software and firmware relevant to Apple Pay and Apple Cash.

4.5. Apple Pay

With Apple Pay, users can enroll credit cards and debit cards to perform transactions using an iOS mobile device. All transactions and usage scenarios that can be performed with Apple Pay are detailed in [AP].

Security Note: The User SHALL NEVER perform Apple Pay card provisioning on a device that is plugged into another piece of equipment.

4.6. Apple Cash

Apple Cash allows several different operations, including payments and transfer of money from a debit card to Apple Cash. All aspects related to Apple Cash are documented in [APC].

4.7. Operational failures

Two guides [TOUCHID_ISSUE] and [DISABLE] are provided for handling the device in cases where:

- Touch ID does not work
- The User forgets the passcode
- The device is disabled

4.8. Security updates, announces and registering

[SEC-ANNOUNCE] allows any user to sign up to be notified about security issues and updates.

[SEC-ISSUE] alerts users about security issues related to their Apple devices and corresponding actions to take.

[SEC-REPORT] provides any person, Apple customer or not, directions to report a security or privacy vulnerability.

[SEC-UPDATE] lists the latest security updates for Apple software products.

Annex A - Card issuer Security Objectives

For Apple Pay services (Apple Pay and Apple Cash), the card issuer or its service provider is the third party in charge of:

- Management of user data for Apple Pay services
- Management of user data for Apple Cash services
- Processing Apple Pay transactions
- Processing Apple Cash transfers

The card issuers authorized to provision cards (for their cardholders, or to the cardholders of their affiliates) enforce the following Security Objectives:

Environment Security Objectives	Description
Cardholder and Apple Pay/Apple Cash Perso	The card issuer is responsible for verifying that the User is authorized to perform a transaction on the payment account linked to the card used as a reference, before allowing the card personalization. The card issuer also ensures the robustness of the personalization data, to prevent attacks like forgery, counterfeiting, or corruption.
Card Data	The card issuer is responsible for using the appropriate security measures to protect the confidentiality and the integrity of the sensitive card data and for guaranteeing the authenticity of the card data during enrolment.
Card Delete	The card issuer of a payment card provisioned on a device is informed after the User removes the card from that device, removes that device from the iCloud account, or performs a device Erase All Content and Settings. The card issuer ensures the provisioned card is removed from the User's payment account (i.e., the unlinking process of the DPAN from the FPAN, which is done by the card issuer or the corresponding TSP).
Apple Pay Transaction Verification	For Apple Pay, the cryptogram released by the Secure Element for an Apple Pay transaction is verified by the card issuer (or its service provider such as the payment network). The cryptogram validation result allows the card issuer to approve or reject the transaction. The payment is invalidated if this verification fails.
Statement	For Apple Pay, the card issuers ensure that the statement associated to the DPAN (list of transactions) is fully accurate and includes, but is not restricted to, the amount and recipient of each transaction. For Apple Cash, the payment card issuer ensures that the ledger associated to an Apple Cash account (list of transfers including completed/canceled/pending) is fully accurate.
Dynamic Linking	For eCommerce transactions, the card issuer (or its service provider) verifies the cryptographic based dynamic linking of the transaction data (including amount and payee). The payment is invalidated if this verification fails.
CDCVM	Payment networks or card issuers are responsible for ensuring that Express transactions can only be accepted for transit-specific use by requiring that non-transit Apple Pay payment transactions have a successful CDCVM.

Annex B - Apple Server Security Objectives

Apple servers are in charge of:

- Management of a User's iCloud account
- Management of User enrollment in Apple Pay
- Management of User enrollment in Apple Cash
- Management of iOS releases
- Device's interface for processing Apple Pay transactions (contact S.Issuer)
- Device's interface for processing Apple Cash transfers (contact S.Issuer)

Apple servers enforce a range of security objectives:

Environment Se- curity Objec- tives	Description
Anti-Replay	The Apple Pay server verifies that each payment (e-Commerce Apple Pay transaction or Apple Cash transfer) is not replayed. The payment is invalidated if this verification fails.
Apple Cash Transaction Veri- fication	The Apple Pay server ensures that no Apple Cash transfer can be executed if the submitted quote (received by the server before the User approves) does not match the transaction data (received by the server once the device completes transfer processing). The modifications that the server is able to detect cover but are not limited to, modification on the amount and the recipient.
Dynamic Linking	For eCommerce transactions, the Apple Pay server preserves the cryptographic based dynamic linking of the transaction data (including amount and payee). The payment is invalidated if this verification fails.
Genuine_Wallet	The Apple Wallet application is provided and signed by Apple.

Annex C – User Security Objectives

The user of the TOE is responsible for:

- Taking appropriate measures to control access to the TOE

The user enforces the following security objective:

Environment Se- curity Objec- tives	Description
User	The User is responsible for ensuring that: • They authorize all the Apple Pay/Cash activities that are performed on the device • The passcode is robust and protected • Only their own biometrics credentials are enrolled (they do not enroll biometrics of someone else)

Change History

Date	Version	Author	Comments
2025-03-17	1.0	Apple	Initial version
2025-04-03	1.1	Apple	Minor updates